

Şirag Erkol

CONTACT INFORMATION	Northwestern University Center for Science of Science and Innovation 1800 Sherman Ave, Evanston, IL 60201	(812) 558-8938 siragerkol@gmail.com siragerkol.github.io
ACADEMIC POSITIONS	Northwestern University , Evanston, IL <i>Research Assistant Professor</i> Northwestern University , Evanston, IL <i>Postdoctoral Fellow</i> Indiana University , Bloomington, IN <i>Research & Teaching Assistant</i> Boğaziçi University , Istanbul, Turkey <i>Research & Teaching Assistant</i>	July 2025 – present July 2023 – June 2025 August 2017 – May 2023 October 2014 – August 2017
EDUCATION	Indiana University , Bloomington, IN <i>Ph.D. in Informatics, Complex Networks and Systems track</i> • Advisor: <i>Assoc. Prof. Filippo Radicchi</i> • Thesis: <i>Identifying Influential Spreaders in Complex Networks</i> • Minor: Computer Science <i>M.S. in Informatics</i> Boğaziçi University , Istanbul, Turkey <i>M.S. in Industrial Engineering</i> • Advisor: <i>Assoc. Prof. Gönenç Yücel</i> • Thesis: <i>Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics</i> <i>B.S. in Industrial Engineering</i> • Graduation Project: <i>Analyzing Parking Stand Requirements for Istanbul Ataturk Airport</i>	August 2017 – May 2023 September 2014 – August 2016 September 2010 – June 2014
WORK EXPERIENCE	Pacific Northwest National Laboratory , Richland, WA <i>Ph.D. Intern</i>	June 2021 – August 2021
HONORS & AWARDS	IU Luddy Summer Research Award (\$4400)	2022
PUBLICATIONS	Published: Symmetry breaking in optimal transport networks S. Patwardhan, M. Barthelemy, Ş. Erkol , S. Fortunato, F. Radicchi, <i>Nature Communications</i> , 15, 3758 (2024) [DOI] [arXiv] Consistency pays off in science Ş. Erkol , S. Sikdar, F. Radicchi, and S. Fortunato, <i>Quantitative Science Studies</i> , 4, 491 (2023) [DOI] [arXiv] [GitHub] [Website] Effective submodularity of influence maximization on temporal networks Ş. Erkol , D. Mazzilli and F. Radicchi, <i>Physical Review E</i> , 106, 034301 (2022) [DOI] [arXiv]	

Who is the best coach of all time? A network-based assessment of the career performance of professional sports coaches

Ş. Erkol and F. Radicchi, *Journal of Complex Networks*, 9, cnab012 (2021)

[DOI] [arXiv] [Website]

Influence maximization on temporal networks

Ş. Erkol, D. Mazzilli and F. Radicchi, *Physical Review E*, 102, 042307 (2020)

[DOI] [arXiv]

Systematic comparison between methods for the detection of influential spreaders in complex networks

Ş. Erkol, C. Castellano and F. Radicchi, *Scientific Reports*, 9, 15095 (2019)

[DOI] [arXiv]

Influence maximization in noisy networks

Ş. Erkol, A. Fageeh and F. Radicchi, *EPL*, 123, 58007 (2018)

[DOI] [arXiv]

Influence Maximization Based on Partial Network Structure Information: A Comparative Analysis on Seed Selection Heuristics

Ş. Erkol and G. Yücel, *International Journal of Modern Physics C*, 28, 1750122 (2017)

[DOI]

PRESENTATIONS

Talks:

Effective submodularity of influence maximization on temporal networks,
NetSci 2022, Online

July 2022

Influence maximization on temporal networks, Networks2021, Online

July 2021

Influence maximization on temporal networks, University of Limerick,
Ireland, Online

March 2021

Posters:

Consistency pays off in science
ICSSI 2023, Evanston, IL, USA

June 2023

Systematic comparison between methods for the detection of influential
spreaders in complex networks
NetSci 2020, Online

September 2020

Influence maximization in noisy networks
NetSci 2019, Burlington, VT, USA

May 2019

COMMUNITY SERVICE

Journal Reviewer: Applied Mathematics and Computation; Chaos, Solitons & Fractals; EPJ Data Science; Expert Systems with Applications; IEEE Transactions on Network Science and Engineering; Journal of Complex Networks; Journal of Computational Science; Pattern Recognition; Physical Review E; Royal Society Open Science; Science Advances

Conference Reviewer: ISSI (2023); ICSSI (2024); IC2S2 (2026)

Journal Subreviewer: PLoS One, PNAS

COMPUTER SKILLS

Frequent user of Python, L^AT_EX
Experience in Django, NetLogo, HTML, CSS
Familiar with MySQL, C, R, Git, Stella

LANGUAGES

English (fluent), Turkish (native), Armenian (native), French (beginner), Spanish (beginner)

Last updated: April 2026